Zhiling Chen

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Education

- **University of Connecticut** — *Aug. 2023 - present*
  Ph.D. student at Mechanical Engineering, advised by Prof. Farhad Imani — Storrs, CT
- **Boston University** — *Sep. 2021 - Aug. 2022*
  Master of Science in Applied Data Analytics — Boston, MA
- **Waterford Institute of Technology** — *Sep. 2020 - June. 2021*
  Bachelor of Science (Honours) in Software Engineering Practice — Waterford, Ireland
- **Nanjing University of Information Science & Technology** — *Sep. 2017 - June. 2021*
  Bachelor of Engineering in Software Engineering — Nanjing, China

Research Interests

My research focuses on **Physical AI and Robot Learning**, with an emphasis on building **agentic robots for high-precision manufacturing**, spanning **agentic robotics, vision-language-action (VLA) models, multimodal reasoning, and recursive self-improvement (RSI)**. My current work explores how robots deployed in **real-world industrial settings** can **recursively self-improve**: diagnosing their own failures, **rapidly acquiring new skills** from deployment experience, and executing **high-precision tasks quickly and reliably**.

Skills

- **Frameworks & languages:** Python/PyTorch, JAX (VLA fine-tuning via openpi, flow-matching policies), LLM API integration (tool calling, structured output, code-as-policy), agentic coding (Codex, Claude Code), ROS2, Linux, Git, Docker
- **Model families:** Qwen-VL, InternVL, LLaVA, GPT, Gemini, CLIP, BLIP2, SigLIP, YOLO, Grounding DINO
- **Robot learning & embodied AI:** VLA (π_0, $\pi_{0.5}$), imitation learning (ACT, Diffusion Policy)
- **Robotics & simulation:** bimanual xArm, UR3, Unitree H2 humanoid; Keyence laser profiler; MuJoCo, Issac Sim
- **Multimodal & retrieval:** vision-language learning, point clouds, depth/3D perception, RAG, CLIP embeddings
- **Data & prototyping:** multimodal robot datasets, benchmark design (ScanBot), data collection pipelines (VR teleoperation), SolidWorks, Blender, OpenSCAD, 3D printing

Publications Google Scholar

First-Author Publications (9)

[1] Zhiling Chen*, Jingzhan Ge, Ruimin Chen, Matthew P. Castanier, David Gorsich, Farhad Imani. **Task-Specified Active Metrological Inspection with Measurement-Steered VLA Manipulation and Deterministic Evidence Gating**. *Under Review*.

[2] Zhiling Chen*, Yang Zhang, Fardin Jalil Piran, Qianyu Zhou, Jiong Tang, Farhad Imani. **ScanBot: A Benchmark for Precision Robotic Surface Scanning with Industrial Laser Profilers**. *IROS 2026*.

[3] Zhiling Chen*, David Gorsich, Matthew P. Castanier, Yang Zhang, Jiong Tang, Farhad Imani. **How Should a Robot Configure Its Laser Scanner for Inspection?**. *IROS 2026*.

[4] Zhiling Chen*, Farhad Imani. **A Multi-Expert Framework for Enhancing Multimodal Large Language Models in Industrial Anomaly Detection**. *Pattern Recognition*.

[5] Zhiling Chen*, Hanning Chen, Mohsen Imani, Farhad Imani. **Can Multimodal Language Model be Guided to Improve Industrial Anomaly Detection?**. *IDETC 2025*.

[6] Zhiling Chen*, Hanning Chen, Mohsen Imani, Ruimin Chen, Farhad Imani. **Vision Language Model for Interpretable and Fine-grained Detection of Safety Compliance in Diverse Workplaces**. *Expert Systems with Applications*.

[7] Zhiling Chen*, Danny Hoang, Fardin Jalil Piran, Ruimin Chen, Farhad Imani. **Federated Hyperdimensional Computing for Hierarchical and Distributed Quality Monitoring in Smart Manufacturing**. *Internet of Things*.

[8] Zhiling Chen*, Danny Hoang, Ruimin Chen, Farhad Imani. **Distributed Hyperdimensional Computing for Real-Time Data Aggregation and Interpretable Quality Monitoring in Manufacturing**. *IMECE 2024*.

[9] Zhiling Chen*, David Gorsich, Matthew P. Castanier, Yang Zhang, Jiong Tang, Farhad Imani. **Task-Aware Scanning Parameter Configuration for Robotic Inspection Using Vision Language Embeddings and Hyperdimensional Computing**. *Under Review*.

Co-Author Publications (5)

[1] Fardin Jalil Piran*, Zhiling Chen, Yang Zhang, Qianyu Zhou, Jiong Tang, Farhad Imani. **Privacy-preserving Federated Learning with Differentially Private Hyperdimensional Computing**. *Computers and Electrical Engineering*.

[2] Qianyu Zhou*, Yang Zhang, Zhiling Chen, Fardin Jalil Piran, Jiong Tang, Farhad Imani. **Few-Shot Vision-Language Models Grounded with Hierarchical Retrieval for Wind Turbine Blade Inspection**. *Results in Engineering*.

[3] Kiarash Naghavi Khanghah*, Zhiling Chen, Lela Romeo, Qian Yang, Rajiv Malhotra, Farhad Imani, Hongyi Xu. **Zero-Shot Anomaly Detection in Laser Powder Bed Fusion Using Multimodal RAG and Large Language Models**. *Journal of Mechanical Design*.

[4] Kiarash Naghavi Khanghah*, Zhiling Chen, Lela Romeo, Qian Yang, Rajiv Malhotra, Farhad Imani, Hongyi Xu. **Multimodal RAG-driven Anomaly Detection and Classification in Laser Powder Bed Fusion using Large Language Models**. *2025 DFMLC Best Paper Award*.

[5] Fardin Jalil Piran*, Zhiling Chen, Yang Zhang, Qianyu Zhou, Jiong Tang, Farhad Imani. **Privacy-Preserving Decentralized Federated Learning via Explainable Adaptive Differential Privacy**. *Under Review*.

Selected Projects

- **FRAME: Task-Specified Active Metrological Inspection** *2026 - Present*
  Tools: Dual xArm 6, Keyence LJ-X8200, RealSense D435i, Qwen3-VL-8B, $\pi_{0.5}$ VLA, LoRA [🌐]
  - Developed **FRAME**, a hierarchical dual-arm robotic system that translates natural-language inspection instructions and structured specifications into closed-loop **look–reorient–scan–verify–sort** execution for high-precision manufacturing inspection.
  - Integrated a **VLA-based manipulation expert** with calibrated laser profilometry, using Qwen3-VL for task grounding and surface correspondence and $\pi_{0.5}$ for part reorientation and sorting, while deterministic evidence gates verify measurement validity and coverage before issuing conformance decisions.
  - Built and evaluated a **real-world precision inspection benchmark** spanning four controlled workpiece families, five inspection-task families, and CNC parts, demonstrating higher end-to-end reliability, fewer false accepts, and faster task completion than exhaustive inspection and direct VLM control.

- **Open Dual-Arm Lab: Bimanual Robotics Platform** *Mar 2026 - now*
  Tools: xArm6 (×2), Meta Quest 3, ROS2, Python, RealSense [🐙]
  - Built an open-source **dual-arm robotic platform** for bimanual manipulation and data collection, integrating two xArm6 manipulators with a unified control and networking system.
  - Developed a **VR-based teleoperation system** using Meta Quest 3, enabling real-time, intuitive control of dual-arm robots via browser-based interfaces.
  - Implemented a **low-latency control pipeline** with websocket communication, motion filtering (One Euro Filter), and safety constraints for stable teleoperation.
  - Designed a **scalable data collection framework** for robot learning, supporting synchronized logging of robot states, actions, and multimodal observations for imitation learning and VLA training.
  - Currently developing a **heterogeneous dual-arm active inspection system**, combining a gripper-equipped manipulator and a laser line scanner arm for task-driven active inspection.

- **ScanBot: Instruction-Conditioned Robotic Surface Scanning Dataset** *Fed 2025 - May 2025*
  Tools: UR3 Robot Arm, Keyence LJ-X8200, RealSense D435i, GoPro, ROS2, PyTorch [🌐]
  - Developed **ScanBot**, the first multimodal dataset for instruction-conditioned robotic surface scanning, covering 12 objects across 6 industrially relevant task types.
  - Implemented a **robotic scanning system** integrating UR3 arm, Keyence LJ-X8200, RealSense D435i, and GoPro to capture synchronized RGB-D, laser profiles, robot states, and third-person video.
  - Created a **comprehensive benchmark** by evaluating state-of-the-art MLLMs (GPT-4.1, OpenAI o3, Gemini 2.5) on perception–planning–execution loops, revealing key limitations in trajectory stability and parameter tuning.

Experience

- **UCONN ISCL Lab [🌐]** *Aug. 2023 - Present*
  Research Assistant Storrs, CT
- **Wuxi CHAOTONG Intelligent Research Institute Co., Ltd** *Jul. 2019 - Sep. 2019*
  Software Engineering Intern Wuxi, China

Honors and Awards

- **Pratt & Whitney Advanced Systems Engineering Fellowship** Aug. 2026
- **ASME DFMLC Best Paper Award** Aug. 2025
- **Pratt & Whitney Advanced Systems Engineering Travel Fellowship** Jan. 2025